

Notes: Bagwell and Staiger (AER, 1999)

An Economic Theory of GATT

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1 Big picture

- **Question.** What is the economic logic behind the core legal principles of GATT: reciprocity and nondiscrimination? The paper asks whether these principles can be understood as simple rules that help governments reach efficient trade agreements.
- **Core answer.** The fundamental inefficiency in unilateral tariff setting is the **terms-of-trade externality**: each government can shift part of the cost of protection onto foreign exporters by changing world prices. GATT rules are useful because they neutralize this cost-shifting channel.
- **Key mechanism.** Reciprocity requires mutual tariff changes that preserve world prices. If world prices are fixed, governments cannot use tariff changes to manipulate the terms of trade. They then behave as if they only care about local-price distortions and political objectives.
- **Role of nondiscrimination.** In a multicountry world, discriminatory tariffs create an additional local-price externality. MFN/nondiscrimination removes this extra channel, so externalities again travel only through world prices, where reciprocity can neutralize them.
- **Main implication.** Reciprocity and nondiscrimination work together. Reciprocity handles world-price externalities; nondiscrimination prevents discriminatory tariff structures from creating local-price externalities.
- **View on preferential agreements.** Free-trade areas and most preferential agreements undermine the multilateral system because they violate nondiscrimination and revive local-price externalities. Customs unions may be compatible only under narrow conditions.

2 Incremental contribution of the paper

- **A unified theory of GATT.** Earlier trade theory emphasized optimal tariffs and trade wars, while legal scholarship described GATT principles. This paper integrates the two by interpreting the legal pillars of GATT as efficiency-enhancing rules.
- **General government preferences.** The model does not require governments to maximize national income only. Government objectives may include political-economy and distributional concerns. This makes the theory robust to political models of protection.
- **Terms-of-trade externality as the central problem.** The paper shows that, within this broad class of preferences, political motives shape desired local prices, but they do not create the international inefficiency. The inefficiency comes from cost shifting through world prices.
- **Politically optimal tariffs.** The paper introduces a useful benchmark: the tariffs governments would choose if they cared about local political and economic objectives but did not value terms-of-trade manipulation. These tariffs are efficient in the two-country model.
- **Reciprocity as an efficiency device.** The paper gives a formal economic interpretation of the GATT practice of exchanging balanced concessions. This is not merely mercantilist bargaining; it is a way to liberalize without changing world prices.
- **Nondiscrimination as a necessary complement.** The paper explains why MFN is not just a fairness principle. It is required to make reciprocity work in a multicountry setting because it ensures that externalities are transmitted only through world prices.

- **Critique of preferential agreements.** The paper provides a theoretical reason why free-trade areas and discriminatory regional arrangements may threaten the multilateral trading system, even if they liberalize trade among members.

3 Model environment

3.1 Two countries and two goods

The baseline environment is a standard two-good, two-country general-equilibrium trade model. The home country naturally imports good x and exports good y ; the foreign country naturally imports good y and exports good x . Production is perfectly competitive, goods are normal in consumption, and tariffs are nonprohibitive.

Let p denote the domestic local relative price of good x in terms of good y , p^* the foreign local relative price, and p^w the untaxed world relative price. With gross tariffs $\tau = 1 + t$ and $\tau^* = 1 + t^*$,

$$p = \tau p^w, \quad p^* = \frac{p^w}{\tau^*}.$$

The domestic terms of trade are inversely related to p^w , while the foreign terms of trade move with p^w .

3.2 Trade balance and market clearing

Domestic imports of x are

$$M_x(p, p^w) = C_x(p, p^w) - Q_x(p),$$

and domestic exports of y are

$$E_y(p, p^w) = Q_y(p) - C_y(p, p^w).$$

Foreign imports and exports are defined analogously. Trade balance and market clearing determine the equilibrium world price $\tilde{p}^w(\tau, \tau^*)$.

The key comparative-static restriction is that a higher home import tariff raises the home local price and lowers the world price:

$$\frac{\partial p}{\partial \tau} > 0, \quad \frac{\partial \tilde{p}^w}{\partial \tau} < 0.$$

Symmetrically, a higher foreign tariff lowers the foreign local price and raises the world price.

3.3 Government objectives

The government payoff is written as a function of local and world prices:

$$W(p, p^w), \quad W^*(p^*, p^w).$$

This notation is important because it separates two effects of tariffs:

- the **local-price effect**, which captures domestic economic and political objectives;
- the **world-price effect**, which captures the terms-of-trade externality.

The only structural assumption on government preferences is that each government likes an improvement in its terms of trade holding its local price fixed:

$$W_{p^w} < 0, \quad W_{p^w}^* > 0.$$

This formulation includes national-income maximization and many political-economy models of protection.

4 Unilateral tariffs and the purpose of trade agreements

4.1 Nash tariffs

Under unilateral tariff setting, each government maximizes its own payoff taking the other government's tariff as given. The first-order conditions are

$$\begin{aligned} W_p \frac{\partial p}{\partial \tau} + W_{p^w} \frac{\partial p^w}{\partial \tau} &= 0, \\ W_{p^*} \frac{\partial p^*}{\partial \tau^*} + W_{p^w}^* \frac{\partial p^w}{\partial \tau^*} &= 0. \end{aligned}$$

Define

$$\lambda = \frac{\partial p^w / \partial \tau}{\partial p / \partial \tau} < 0, \quad \lambda^* = \frac{\partial p^w / \partial \tau^*}{\partial p^* / \partial \tau^*} < 0.$$

Then the Nash conditions can be written as

$$W_p + \lambda W_{p^w} = 0, \quad W_{p^*} + \lambda^* W_{p^w}^* = 0.$$

Because $\lambda W_{p^w} > 0$ for the home country, the home government chooses a tariff that is too high from a joint-efficiency perspective. It ignores the loss imposed on foreign exporters through the lower world price.

4.2 Proposition 1: Nash tariffs are inefficient

The Nash equilibrium is inefficient because each government uses tariffs partly to improve its terms of trade. This creates excessive trade restriction. The paper's first major result is therefore:

Nash equilibrium tariffs are inefficient.

The inefficiency is not caused by political preferences per se. Political preferences affect where governments want local prices to be, but the strategic inefficiency comes from the world-price effect.

4.3 Proposition 2: reciprocal trade agreements require liberalization

Starting from Nash tariffs, a mutually beneficial reciprocal trade agreement must involve reciprocal tariff reductions. Since Nash tariffs are too restrictive, both governments can gain by lowering tariffs in a way that controls the distributional consequences of world-price changes.

4.4 Politically optimal tariffs

The paper defines **politically optimal tariffs** as the tariffs governments would choose if they cared about local-price effects but did not value the terms-of-trade consequences of their policies:

$$W_p = 0, \quad W_{p^*}^* = 0.$$

These tariffs incorporate political-economy motivations, but remove cost-shifting incentives.

4.5 Proposition 3: politically optimal tariffs are efficient

Politically optimal tariffs are efficient. The intuition is direct: if governments ignore world-price manipulation, they choose tariffs based only on domestic local-price objectives. Once both governments do this, the only cross-country inefficiency has been removed.

In the special case of national-income-maximizing governments, politically optimal tariffs correspond to reciprocal free trade.

5 Reciprocity

5.1 Definition of reciprocity

The paper interprets reciprocity as mutual changes in trade policy that generate equal changes in import volumes across trading partners. In the two-country model, this condition implies that world prices remain unchanged.

For tariff changes from (τ^0, τ^{*0}) to (τ^1, τ^{*1}) , reciprocity requires that changes in import volumes are balanced at the original world price. Using balanced trade, the condition reduces to

$$[p^{w1} - p^{w0}]M_x(p(\tau^1, p^{w1}), p^{w1}) = 0.$$

With positive trade, this implies

$$p^{w1} = p^{w0}.$$

Thus, reciprocity neutralizes the terms-of-trade channel.

5.2 Reciprocity and the balance of concessions

The paper explains why governments in GATT negotiations seek a balance of concessions. From a standard free-trade perspective, this can look like mercantilism: governments seem to treat exports as good, imports as bad, and balanced trade expansion as desirable. Bagwell and Staiger show that this behavior has an economic logic.

At the Nash equilibrium, a government would like its trading partner to reduce tariffs because this improves its terms of trade. But it does not want to liberalize unilaterally because unilateral liberalization worsens its own terms of trade. A reciprocal tariff reduction solves this problem by expanding trade while holding world prices fixed.

5.3 Proposition 4: reciprocal liberalization raises welfare near Nash

Starting from Nash tariffs, reciprocal trade liberalization that conforms to reciprocity raises each government's welfare monotonically until one government reaches its preferred local price at the original world price:

$$\min[W_p, W_p^*] = 0.$$

If countries are symmetric, this path leads exactly to the politically optimal outcome.

5.4 Renegotiation under reciprocity

GATT also uses reciprocity in renegotiation. If one country raises a previously bound tariff, trading partners are allowed to withdraw substantially equivalent concessions. The paper models this as a process in which renegotiation preserves the original world price.

This creates a strong selection result. Any efficient tariff pair that is not politically optimal will be renegotiated, because at the prevailing world price at least one government would prefer a different local price. The only efficient tariffs that are immune to such renegotiation are politically optimal tariffs.

5.5 Proposition 5: efficient agreements under reciprocity

The paper's key result for reciprocity is:

An efficient trade agreement can be implemented under reciprocity if and only if tariffs are set at their politically optimal levels.

Thus reciprocity does more than permit liberalization; it directs negotiations toward a specific point on the efficiency frontier.

6 Nondiscrimination

6.1 Why multicountry trade changes the problem

In a multicountry world, the home country may import from multiple foreign suppliers. If it sets discriminatory tariffs, it may prefer that a larger share of its imports come from the exporter on which it imposes the highest tariff. This creates a new **local-price externality**: foreign countries' local prices affect bilateral trade shares, which affect the home country's tariff revenue.

This externality does not exist in the two-country model. It also disappears when the home country follows MFN, because all foreign suppliers face the same tariff and the home country's terms of trade are summarized by a common world price.

6.2 Politically optimal tariffs and MFN

In the multicountry model, politically optimal tariffs satisfy

$$W_p = 0, \quad W_{p^*j}^{*j} = 0 \quad \text{for all foreign countries } j.$$

But there are generally many politically optimal discriminatory tariff vectors. The paper proves that political optimality is efficient only when it is combined with MFN.

6.3 Proposition 6: political optimality requires MFN for efficiency

The central nondiscrimination result is:

Politically optimal tariffs are efficient if and only if they conform to MFN.

The reason is that MFN channels all international externalities through world prices. Without MFN, discriminatory tariffs allow externalities to travel through local prices and trade shares.

6.4 Proposition 7: reciprocity and nondiscrimination together

When the multicountry renegotiation game is considered, the paper obtains the core joint result:

An efficient multilateral trade agreement can be implemented under reciprocity if and only if tariffs conform to MFN and are set at politically optimal levels.

This is the paper's formal explanation of why reciprocity and nondiscrimination are complementary pillars of GATT.

7 Preferential agreements

7.1 Free-trade areas

A free-trade area eliminates internal trade barriers among members but allows each member to keep its own external tariffs. Because this arrangement is discriminatory, it violates MFN and revives the local-price externality.

7.2 Proposition 8: free-trade agreements are incompatible with efficient multilateralism

The paper shows:

An efficient multilateral trade agreement cannot be implemented under reciprocity in the presence of a free-trade agreement.

This does not mean that a free-trade area never raises trade or welfare for members. It means that the GATT mechanism of reciprocity plus nondiscrimination no longer guarantees efficient multilateral outcomes once discriminatory preferences are introduced.

7.3 Customs unions

A customs union eliminates internal barriers and adopts a common external tariff. This can sometimes be treated as a single country in the model. However, for the customs union to be compatible with multilateral efficiency, its members must be **natural integration partners**: their internal allocation of tariff revenue and common local price must be consistent with internal efficiency.

7.4 Proposition 9: customs unions are compatible only under narrow conditions

The paper's customs-union result is:

An efficient multilateral trade agreement can be implemented under reciprocity in the presence of a customs union if and only if the customs-union members are natural integration partners and all external tariffs conform to MFN and political optimality.

This is a narrow exception. The broader message is that preferential agreements generally threaten the efficiency properties of the multilateral system.

8 Rules-based versus power-based bargaining

8.1 Why a rules-based institution?

The paper contrasts GATT's rules-based approach with a power-based bargaining approach. A power-based bargain may reach the efficiency frontier, but the resulting distribution of gains depends heavily on bargaining power. A rules-based approach based on reciprocity reduces the role of power asymmetries and steers negotiations toward the political optimum.

8.2 Participation of weaker countries

The paper argues that powerful countries may support a rules-based institution because it encourages participation by weaker countries. If weaker countries fear exploitation in power-based negotiations, they may refuse to participate. Reciprocity moderates the lawful response of stronger countries and makes participation more attractive to weaker countries.

This argument is especially relevant when private agents make irreversible investments in anticipation of trade negotiations. If negotiations fail, the weaker country may be worse off after such investments are sunk. Rules that reduce bargaining-power asymmetries can help solve this participation problem.

9 Empirical relevance and caveats

9.1 Terms-of-trade manipulation must matter

For the theory to be empirically relevant, governments must be able to affect terms of trade and the resulting cost shifting must be quantitatively meaningful. The paper discusses evidence that

tariff changes and trade restrictions are not fully passed through to domestic consumers, implying that foreign exporters bear part of the cost.

9.2 Evidence discussed in the paper

The authors mention several empirical findings:

- Kreinin's study of GATT negotiations suggests that less than one-third of U.S. tariff concessions were passed to U.S. consumers as reduced import prices, while more than two-thirds accrued to foreign suppliers.
- The exchange-rate pass-through literature also supports the idea that changes in trade costs are not fully passed through.
- Berry, Levinsohn, and Pakes' work on U.S.-Japan automobile VERs shows that world-price effects can substantially change the national cost of trade intervention.

9.3 Caveats

The paper emphasizes two caveats.

- **Enforcement.** The analysis abstracts from enforcement constraints. In practice, imperfect enforcement may prevent governments from reaching the efficiency frontier.
- **Other motives.** The model does not capture every reason governments form trade agreements. Regional agreements may be motivated by security, political stability, or domestic commitment motives that are outside the local-price/world-price framework.

10 Figures

10.1 Figure 1: The World- and Local-Price Effects of a Tariff Change

Read:

- The figure decomposes a tariff increase into a local-price effect and a world-price effect.
- Moving along the iso-local-price locus isolates the world-price/terms-of-trade effect.
- Moving along the iso-world-price locus isolates the local-price effect.
- The figure is the visual foundation for separating political motives from terms-of-trade manipulation.

Economic message: A tariff affects both domestic relative prices and the international terms of trade. The inefficiency in unilateral policy comes from the world-price movement, not from domestic political preferences by themselves.

10.2 Figure 2: The Purpose of a Reciprocal Trade Agreement

Read:

- The Nash point N lies away from the efficiency locus because unilateral tariffs are too restrictive.
- The politically optimal point PO lies on the efficiency locus.
- The contract curve is the part of the efficiency locus that gives both governments gains relative to Nash.
- A reciprocal trade agreement moves governments from inefficient Nash tariffs toward lower and more efficient tariffs.

Economic message: Trade agreements exist to replace inefficient Nash tariffs with lower tariffs on the contract curve. The political optimum is the point governments would choose without terms-of-trade manipulation.

10.3 Figure 3A: Liberalization According to Reciprocity - The Symmetric Case

Read:

- In the symmetric case, the iso-world-price locus through the Nash point also passes through the politically optimal point.
- Reciprocal liberalization from Nash keeps the world price fixed.
- Both governments gain until they reach PO .

Economic message: When countries are symmetric, reciprocity can guide tariff reductions all the way from Nash to the politically optimal and efficient outcome.

10.4 Figure 3B: Liberalization According to Reciprocity - The Asymmetric Case

Read:

- In the asymmetric case, the Nash iso-world-price locus need not intersect PO .
- Reciprocal liberalization still raises both governments' welfare at first.
- The process stops at point Z when one government reaches its preferred local price at the unchanged world price.

Economic message: Reciprocity is still efficiency-enhancing near Nash, but asymmetry may prevent informal reciprocal liberalization from reaching the political optimum without further bargaining.

10.5 Figure 4: Renegotiation Under Reciprocity

Read:

- Points A and B are efficient but not politically optimal.
- At each of these points, one government would prefer to move along the same iso-world-price locus to a different local price.
- Reciprocity permits renegotiation that preserves the world price.

- Only PO is both efficient and immune to renegotiation.

Economic message: GATT-style reciprocity in renegotiation selects the politically optimal point from the broader efficiency frontier.

10.6 Figure 5A: The Locus of Tariffs Implementable Under Reciprocity

Read:

- The figure shows the set of tariff pairs that can be implemented under the reciprocity-constrained negotiation game.
- The implementable locus moves through $R \rightarrow PO \rightarrow R$.
- Reciprocity shrinks the feasible bargaining set relative to unconstrained bargaining.

Economic message: Reciprocity limits bargaining outcomes that move too far from political optimality and thereby steers negotiations toward PO .

10.7 Figure 5B: The Feasible Set of Bargaining Outcomes Implementable Under Reciprocity

Read:

- The dashed curve is the unconstrained efficiency frontier.
- The bold curve is the welfare frontier implementable under reciprocity.
- A power-based outcome may lie at A , while a rules-based reciprocity outcome may lie closer to PO at B .
- The figure also illustrates the participation logic for weaker countries.

Economic message: Reciprocity narrows the bargaining set and reduces the role of power asymmetries, which can make participation more attractive for weaker countries.

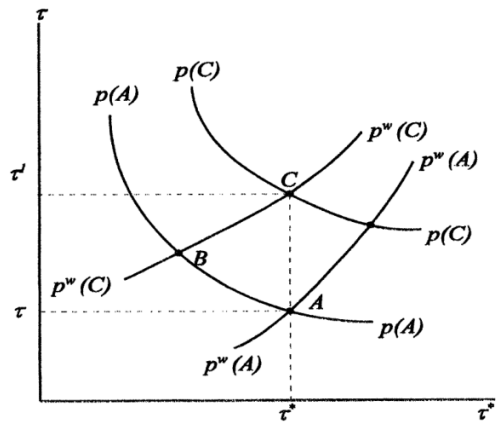


FIGURE 1. THE WORLD- AND LOCAL-PRICE EFFECTS OF A TARIFF CHANGE

(a) Figure 1: World- and local-price effects

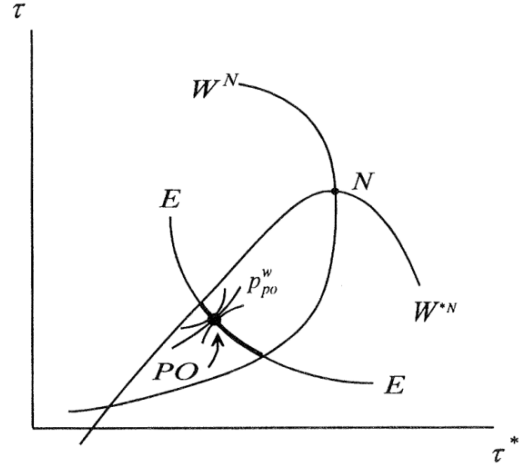


FIGURE 2. THE PURPOSE OF A RECIPROCAL TRADE AGREEMENT

(b) Figure 2: Purpose of a reciprocal trade agreement

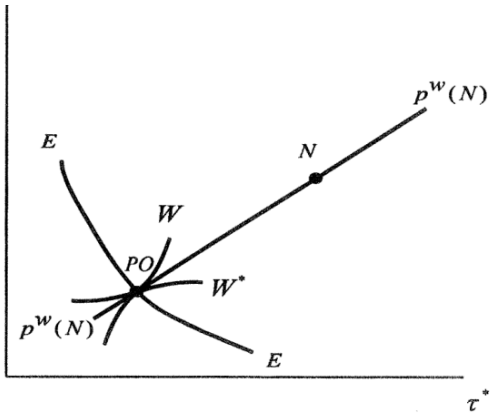


FIGURE 3A. LIBERALIZATION ACCORDING TO RECIPROCITY — THE SYMMETRIC CASE

(a) Figure 3A: Symmetric case

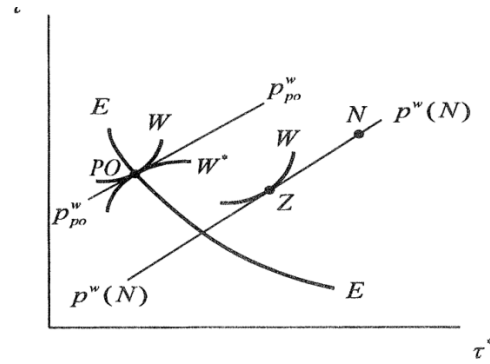


FIGURE 3B. LIBERALIZATION ACCORDING TO RECIPROCITY — THE ASYMMETRIC CASE

(b) Figure 3B: Asymmetric case

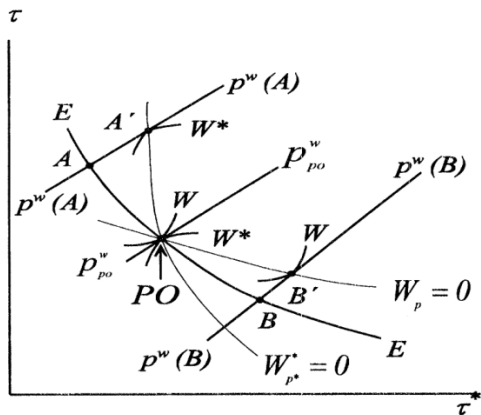


FIGURE 4. RENEGOTIATION UNDER RECIPROCITY

(a) Figure 4: Renegotiation under reciprocity

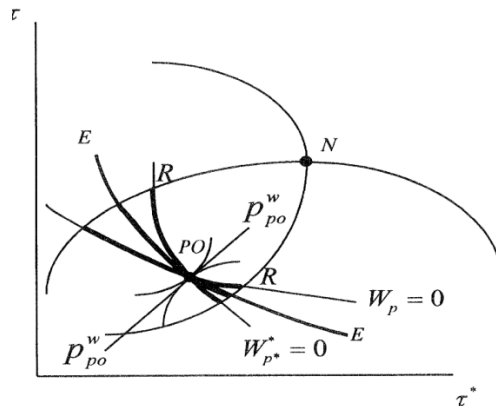


FIGURE 5A. THE LOCUS OF TARIFFS IMPLEMENTABLE UNDER RECIPROCITY

(b) Figure 5A: Tariffs implementable under reciprocity

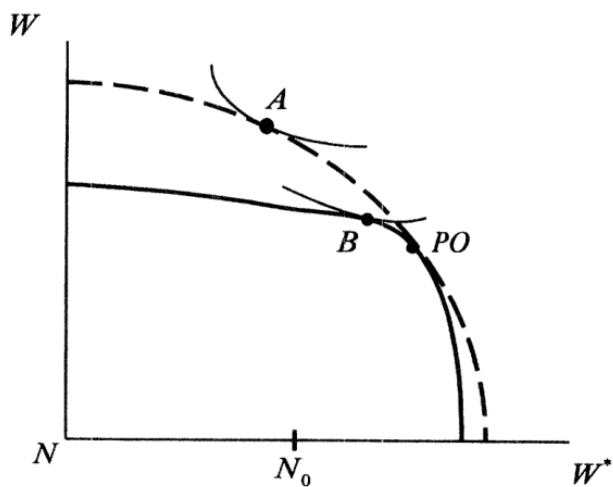


FIGURE 5B. THE FEASIBLE SET OF BARGAINING OUTCOMES IMPLEMENTABLE UNDER RECIPROCITY

Figure 5B: Feasible bargaining outcomes under reciprocity

11 Short summary

- The paper explains GATT as an institution designed to solve the terms-of-trade externality in unilateral tariff setting.
- Governments may have political motives, but the inefficiency that trade agreements solve comes from cost shifting through world prices.

- Politically optimal tariffs are the tariffs governments would choose if they cared about domestic local-price effects but not terms-of-trade manipulation.
- In the two-country model, politically optimal tariffs are efficient.
- Reciprocity preserves world prices, so it neutralizes the terms-of-trade externality.
- Under GATT-style renegotiation, efficient agreements are implementable under reciprocity only at politically optimal tariffs.
- In a multicountry model, nondiscrimination/MFN is necessary because discriminatory tariffs create local-price externalities.
- Reciprocity and nondiscrimination jointly implement politically optimal MFN tariffs.
- Free-trade areas are generally incompatible with this mechanism; customs unions are compatible only under narrow conditions.
- GATT's rules-based structure can encourage participation by weaker countries by reducing the role of bargaining-power asymmetries.

12 Conclusion

12.1 Core conclusion

Bagwell and Staiger provide an economic theory of GATT in which reciprocity and nondiscrimination are not arbitrary legal norms. They are simple institutional rules that help governments undo the terms-of-trade inefficiency created by unilateral tariffs. Reciprocity fixes world prices during negotiations; nondiscrimination ensures that world prices are the only channel through which international policy externalities travel.

12.2 Economic intuition

When governments set tariffs unilaterally, each government has an incentive to use tariffs to improve its own terms of trade. This shifts part of the cost of protection onto trading partners and leads to excessive protection. A trade agreement is useful because it removes this cost-shifting motive. Reciprocity does so by preserving world prices. MFN does so by preventing discriminatory tariffs from creating additional externalities through trade shares and local prices.

12.3 Why this paper matters

The paper is important because it turns GATT's legal architecture into an economic mechanism. The two pillars of the multilateral system are shown to be complementary tools for efficiency. The analysis also explains why preferential agreements may threaten multilateral cooperation: they violate nondiscrimination and therefore weaken the ability of reciprocity to guide governments toward efficient outcomes.

12.4 Limitations

The paper abstracts from enforcement constraints, from domestic commitment motives, and from non-economic objectives such as security or political integration. These omissions are important, especially for understanding why regional agreements may be tolerated in practice. Still, within the local-price/world-price framework, the paper offers a powerful explanation of why GATT's central rules can improve global trade-policy outcomes.

12.5 Takeaway

The enduring takeaway is that GATT's reciprocity and nondiscrimination rules work together to discipline the terms-of-trade motive in tariff policy. Reciprocity removes the incentive to manipulate world prices; nondiscrimination ensures that there are no additional discriminatory externalities. Together they direct governments toward politically optimal MFN tariffs and provide an economic foundation for the multilateral trading system.